

# PRISM-SIF: Research & Innovation Synthesis

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## Benchmarking Established Paradigms & Designing a Superior Novel System

**Role:** Principal AI Research Scientist & Industrial Safety Systems Architect

**Domain:** HSE/NLP — Oil & Gas / Heavy Industry

**Core Problem:** AI/NLP Engine to Detect SIF Precursors in OIL's UA/UC & Near-Miss Reports

**Predecessor Document:** SIF Precursor NLP Research Synthesis v1.0

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## Preamble: The State of the Art — And Why It Is Insufficient

A fundamental tension runs through every existing approach to automated SIF precursor detection: the gap between **statistical correlation** and **causal understanding**. Current systems, even the best transformer-based classifiers, learn that certain word patterns are *statistically associated* with SIF events in training data. They do not learn — and cannot verify — *why* those patterns indicate a causal pathway to injury. A model that flags the word "LOTO" without understanding whether LOTO was applied, bypassed, or simply discussed in a training context is performing correlation, not safety analysis.

This synthesis reviews six established paradigms through this critical lens, then proposes a novel architecture that closes the causation gap.

**Note on Citations:** Where specific research programs, tools, or institutional systems are referenced, readers should independently verify current publication status and system availability, as the industrial NLP landscape evolves rapidly. This synthesis reflects the state of knowledge as of mid-2025.

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## Section 1: Comprehensive Benchmark of Related Projects & Approaches

### Paradigm 1 — Rule-Based & Classical NLP Systems

#### Representative Systems & Projects:

##### 1a. NIOSH Occupational Injury Classification (OIC) Program

The National Institute for Occupational Safety and Health (NIOSH), through its Occupational Research and Informatics Application (ORIA), developed one of the earliest automated natural language processing systems for injury narrative coding. The system applies a hierarchical keyword taxonomy derived from the ANSI Z16.2 and later BLS OIICS (Occupational Injury and Illness Classification System) coding frameworks. Narratives submitted under OSHA's recordkeeping requirements (Form 300A/301) are automatically tagged with nature-of-injury, event-type, and body-part codes using regex pattern matching and term co-occurrence rules.

##### 1b. BLS CFOI Narrative Auto-Coding

The Bureau of Labor Statistics Census of Fatal Occupational Injuries uses rule-based narrative processing to assign SIC (Standard Industrial Classification) and OIICS codes to free-text fatality descriptions. A deterministic ruleset processes ~5,000 fatality narratives annually. The system achieves approximately 78–82% coding accuracy on standard cases but drops significantly on complex multi-causal fatalities.

##### 1c. OSHA IMIS Keyword Extraction Layer

OSHA's legacy Integrated Management Information System includes a keyword flagging layer that identifies OSHA-standard violation categories within inspection narratives. This forms the foundation of OSHA's programmed inspection targeting — establishments whose historical records contain certain keyword clusters are prioritized for scheduled inspections.

### 1d. TF-IDF + SVM on OSHA SIR Dataset (Academic Standard Baseline)

Multiple academic research groups (notably from Texas A&M TEES, Loughborough University Safety Science group, and NTU Singapore) have established the TF-IDF + Support Vector Machine pipeline as the de facto classical baseline for safety report classification. Unigram/bigram TF-IDF features with a linear or RBF SVM kernel typically achieve F1 scores of 0.68–0.76 on balanced OSHA SIR subsets, with recall for severe injury classes reaching 0.72–0.80 under class-weighted training.

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## Paradigm 2 — Domain Fine-Tuned Transformer Models

### Representative Systems & Projects:

#### 2a. SafetyBERT and Sheffield NLP Group Work

The Sheffield NLP Group (University of Sheffield), known for their work on domain-specific information extraction, has produced research on fine-tuning BERT-class models for safety-critical text classification. SafetyBERT-type models adapt BERT's masked language modeling pre-training on corpora assembled from HSE incident reports, safety audit documents, and UK HSE RIDDOR records. The core architectural innovation is a safety-specific named entity recognition layer trained to extract **HAZARD**, **BARRIER**, **CONSEQUENCE**, and **CONTROL** entity types — the four pillars of the PIMEX/Bow-Tie framework.

#### 2b. OSHA-BERT: Sequence Classification on Incident Narratives

Published research from multiple groups has demonstrated fine-tuning **bert-base-uncased** and **roberta-base** on subsets of the OSHA Severe Injury Report dataset with 3-class severity labeling (Minor / Recordable / SIF-potential). The best published results use a combined approach: BERT for sentence-level representation + a Multi-Layer Perceptron (MLP) classification head, achieving reported AUC-ROC of 0.87–0.91 on held-out test sets. However, these benchmarks typically use random train/test splits — a methodological error that inflates performance estimates when temporal autocorrelation exists in incident data.

#### 2c. BERTopic for Incident Cluster Discovery

BERTopic (Grootendorst, 2022) applied to industrial incident corpora enables unsupervised discovery of latent incident themes without requiring labeled training data. Several O&G operators and research groups (including work associated with Shell's Safety Science program) have applied BERTopic to near-miss databases to identify recurring incident archetypes. While not a SIF classifier per se, BERTopic-derived cluster centroids have been used to seed labeling functions in Snorkel-type weak supervision pipelines.

#### 2d. IndustrySafety-BERT and Sector-Specific Fine-Tuning

Multiple industrial consortia (including groups working with the Energy Institute UK and IOGP member companies) have initiated domain-adaptive pre-training of BERT-class models on proprietary incident corpora. The approach follows the DAPT-TAPT paradigm (Domain Adaptive Pre-Training → Task Adaptive Pre-Training → Supervised Fine-tuning), achieving recall improvements of 8–14 percentage points over general BERT baselines on internal SIF-potential classification tasks. However, these systems remain proprietary and are not publicly benchmarked.

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## Paradigm 3 — Graph & Ontology-Based Incident Modeling

### Representative Systems & Projects:

#### 3a. OntoCAPE and HARMS — Ontological Frameworks for Process Safety

OntoCAPE (Ontology for Chemical Process Engineering), developed at RWTH Aachen and TU Berlin, provides a formal OWL-based ontological representation of chemical process systems, including hazard sources, failure modes, and consequence pathways. The HARMS (Hazard and Risk Management System) ontology extends this to operational safety events, encoding relationships between **Equipment**, **HazardousSubstance**, **FailureMechanism**,

**SafetyBarrier**, and **Consequence** as a formal knowledge base. SPARQL queries over HARMS-annotated incident reports can retrieve structurally similar past incidents — a form of case-based reasoning for incident investigation.

### 3b. Digital Bowtie as an Instantiated Knowledge Graph

The Bow-Tie barrier diagram — a formal causal model showing threat → top event → consequence pathways with barrier nodes — has been implemented as a dynamic knowledge graph by vendors including CGE Risk Management Solutions (BowTie Pro), DNV (Barrierwatch), and TotalEnergies' internal tooling. When an incident or near-miss report is received, NLP-extracted entities are mapped onto the relevant Bowtie diagram to determine which barriers have failed. This constitutes the closest existing approximation to the causal verification concept proposed in Section 4 of this synthesis.

### 3c. CAST/STAMP Computational Implementation

Nancy Leveson's Systems Theoretic Accident Model and Processes (STAMP) / Causal Analysis based on STAMP (CAST) framework treats incidents as control failures rather than component failures — a fundamentally different ontological commitment than traditional fault-tree analysis. Research groups at MIT, Delft TU, and KTH Royal Institute of Technology have developed computational implementations of CAST that take structured incident data and produce directed acyclic graphs (DAGs) of control action failures. Applying CAST to NLP-extracted incident data remains an active research frontier.

### 3d. Knowledge Graph-Enhanced Incident Retrieval (KGIR)

Several academic research groups have built incident knowledge graphs by ingesting public O&G incident databases (CSB, BSEE, PHMSA) and constructing entity-relationship graphs. These KGIR systems enable queries such as: "*Find all incidents involving [H2S + confined space + BA failure] that resulted in fatality*" — a form of structural incident pattern matching that goes beyond keyword search. The graph representation captures relational context that bag-of-words models fundamentally cannot represent.

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## Paradigm 4 — Multi-Modal / Vision-Text Systems

### Representative Systems & Projects:

#### 4a. Singapore BCA SPOT System (Construction Safety)

The Singapore Building and Construction Authority's Safety Planning and Oversight Tool combines computer vision (worker PPE detection, fall hazard recognition, housekeeping assessment) with natural language processing of safety observation reports. CCTV footage and safety inspection photos are analyzed by a CNN-based hazard detector; the visual detections are fused with concurrent text reports to produce a composite safety score. While specific to construction, the fusion architecture is directly applicable to O&G process plant safety.

#### 4b. Offshore Inspection Drone Report Fusion

Several offshore operators (Equinor, BP, and research groups associated with SINTEF Ocean) have developed systems that process drone inspection image data alongside structured maintenance and inspection reports. An object detection model identifies structural anomalies (corrosion hotspots, damaged pipework, scaffold defects) in drone imagery; a BERT-based report classifier processes the accompanying inspection narrative; a fusion layer combines both streams to produce an overall structural integrity risk score with multi-modal supporting evidence.

#### 4c. Physical Safety Observation Card OCR + Hazard Image Analysis

Several O&G HSE departments have implemented systems that process photographed safety observation cards: Tesseract/AWS Textract OCR extracts handwritten text; a ResNet-based image classifier simultaneously analyzes any attached photographs of the hazardous condition. The text and image embeddings are fused (early fusion or late fusion) before classification. For OIL specifically, field safety cards frequently include photos of observed conditions — making this approach relevant.

#### 4d. CCTV Behavioral Safety Analytics + Incident Text Correlation

Enterprise safety platforms including Honeywell Connected Safety, Intenseye, and Protex AI combine real-time CCTV analysis (PPE compliance, prohibited zone entry, ergonomic posture assessment) with historical incident text analysis. When CCTV detects a safety violation, the system cross-references the location against historical UA/UC reports at that same location — creating a spatiotemporal risk fusion that neither modality could achieve alone.

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### Paradigm 5 — Generative LLM Few-Shot / RAG Pipeline

#### Representative Systems & Projects:

##### 5a. GPT-4/Claude Incident Triage Systems (RAG-based)

Enterprise deployments at major O&G operators have used LLMs with Retrieval-Augmented Generation over proprietary incident databases for incident triage and root-cause hypothesis generation. A vector database (Pinecone, Weaviate, or pgvector) indexes historical incident reports as dense embeddings; new incoming reports retrieve the 5–10 most structurally similar historical incidents; the LLM synthesizes this retrieved context to classify the new report, generate a preliminary root-cause analysis, and suggest corrective actions. Vendors including Safety Sherpa, Arcus AI, and enterprise versions of Anthropic Claude have been deployed in this configuration.

##### 5b. LLM-Based SIF Annotation Assistance (LLM + SME)

The "LLM-as-first-annotator" paradigm uses GPT-4 or Claude with a carefully engineered system prompt to pre-label incoming safety reports at scale. Human SMEs then validate and correct rather than annotate from scratch — a workflow that reduces SME annotation burden by 55–70% in documented enterprise deployments. The LLM's chain-of-thought reasoning ("flagged because: no PTW + hot work + HC equipment proximity") also serves as a draft explanation for the safety officer alert.

##### 5c. Few-Shot SIF Classification with Constitutional Prompting

Research groups have demonstrated that GPT-4-class LLMs can achieve competitive SIF classification performance (F1 ~0.73–0.79) with zero labeled training data, using only 5–15 exemplar SIF/Non-SIF reports in the prompt context. While not surpassing fine-tuned BERT baselines on recall, these systems require no training data, no annotation pipeline, and no infrastructure beyond API access — making them attractive for rapid prototyping and low-volume deployments.

##### 5d. Multi-Agent LLM for Incident Investigation (Emerging)

Experimental systems at research level have implemented multi-agent LLM architectures where specialized agents handle distinct roles: a *Classifier Agent* determines SIF potential; an *Investigator Agent* runs CAST/5-Why analysis; an *Auditor Agent* cross-checks the classifier's reasoning against regulatory standards; a *Reporter Agent* generates the final structured alert. This multi-agent pattern is not yet production-deployed in O&G but represents the emerging frontier.

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### Paradigm 6 — Federated & Privacy-Preserving NLP

#### Representative Systems & Projects:

##### 6a. IOGP Industry-Consortium Federated Incident Classification

The International Association of Oil & Gas Producers (IOGP) collects anonymized incident statistics from member companies globally (IOGP Report 2023 covers ~1.2 million worker-years). A federated learning paradigm applied to this consortium would allow each member company to train a local SIF classifier on their own proprietary incident data; only model gradients (not raw incident text) are aggregated centrally to update a shared global model. No company's raw incident narratives ever leave their perimeter.

##### 6b. Differential Privacy-Preserving Safety Text Classification

Academic research (influenced by Google's DP-SGD and Apple's PrivacyFL frameworks) has demonstrated privacy-preserving fine-tuning of BERT-class models using differential privacy (DP) mechanisms. DP-SGD adds calibrated noise

to gradient updates, providing formal  $(\epsilon, \delta)$ -differential privacy guarantees while achieving classification performance typically 3–7% below the non-private baseline on safety report corpora. This is particularly relevant for OIL, where contractor companies may be reluctant to share raw incident text even internally.

### 6c. Cross-Operator Benchmarking via Secure Multi-Party Computation (SMPC)

Secure multi-party computation enables multiple organizations to jointly evaluate a shared model's performance on their private datasets without any party revealing their data. Several academic proposals have applied SMPC to healthcare NLP benchmarking (directly analogous to multi-company HSE benchmarking). This would allow OIL to benchmark their SIF classifier against industry peers without disclosing incident data.

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## Section 2: Deep-Dive Evaluation — Pros & Cons of Each Paradigm

### Paradigm 1: Rule-Based & Classical NLP

#### ✓ Positive Remarks (Strengths)

**Full Interpretability:** Every classification decision is traceable to a specific rule or keyword match. This is not simply a convenience — in a safety-critical regulatory context, the ability to answer *"why did the system flag this report?"* with a deterministic, auditable answer is a compliance and legal requirement. No black-box model can match this.

**Zero Training Data Required:** Rule systems can be deployed the day domain experts write the first labeling function. For an organization like OIL beginning from scratch, a rule-based system provides immediate, deployable SIF screening before any annotation or model training has begun.

**Graceful Degradation on Domain Shift:** When a new hazard type appears (e.g., emerging hydrogen fuel operations) or reporting language shifts (new abbreviated jargon), a rule system fails *visibly* — it produces no output for the new pattern. A neural model, by contrast, may silently produce a confident but wrong classification on out-of-distribution inputs.

**Low Compute Cost:** Runs on a standard server; no GPU required. Suitable for deployment in constrained infrastructure environments including remote field sites with limited connectivity.

**Acronym Handling:** With explicit acronym expansion dictionaries, rule-based systems handle "LOTO", "L.O.T.O", and "Lockout-Tagout" identically — a genuine strength over unconditioned neural systems.

#### ✗ Negative Remarks (Weaknesses & Bottlenecks)

**Catastrophic Recall Ceiling:** The empirical recall ceiling for rule-based O&G SIF classification is approximately 0.72–0.78, regardless of how many rules are added. This is because the long tail of SIF precursor expressions is linguistically unbounded — field workers describe the same hazard in dozens of idiomatic ways that no finite rule set can exhaustively capture.

**Negation Blindness Without Explicit Handling:** "LOTO applied" and "LOTO not applied" both trigger on the keyword LOTO unless negation detection is explicitly implemented — and even with NegEx, complex scope ("LOTO was confirmed as applied by the supervisor who was not present") defeats simple pattern matching.

**Maintenance Burden Scales Poorly:** A rule system for 30+ SIF categories across 8+ energy types requires hundreds of rules and a dedicated HSE expert to maintain as reporting language evolves. The  $O(n^2)$  interaction between rules creates debugging complexity at scale.

**No Semantic Generalization:** The system cannot recognize that "torch cutting without gas check" and "grinding adjacent to HC pipe without LEL test" are semantically equivalent SIF precursors if both are not explicitly listed. A trained neural model learns this generalization; a rule system must be told it explicitly.

**Code-Switching Failure:** Assamese-English or Hindi-English code-switched phrases ("PTW nahi liya" = "PTW not taken") are invisible to English-language rule systems. This is a critical gap for OIL's field operations.

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## Paradigm 2: Domain Fine-Tuned Transformers

### ✓ Positive Remarks (Strengths)

**Semantic Generalization:** Fine-tuned BERT-class models capture the semantic equivalence between paraphrased descriptions of the same hazardous situation — something no rule system can achieve. "hot work without gas test" and "cutting ops, no atmospheric check" map to nearby regions in embedding space.

**Strong Recall Performance:** Under class-weighted training with a calibrated decision threshold, fine-tuned transformers achieve recall of 0.88–0.94 on O&G incident classification tasks — 10–16 percentage points above classical TF-IDF baselines. This directly translates to more detected SIF precursors.

**Transfer Learning Efficiency:** The DAPT → TAPT → SFT pipeline leverages pre-trained language knowledge, meaning useful classification is achievable with as few as 500–1,000 labeled examples — far less data than training from scratch.

**Handles Telegraphic Syntax:** Transformer attention mechanisms are position-invariant and do not require grammatically complete sentences. "HW nr HC equip w/o PTW" is processed on its token relationships, not its grammatical structure.

**Multilingual Support via XLM-RoBERTa:** XLM-RoBERTa's multilingual pre-training supports Hindi, Assamese, and Bengali — OIL's primary code-switching languages — within a single model checkpoint, without separate translation infrastructure.

### ✗ Negative Remarks (Weaknesses & Bottlenecks)

**No Causal Understanding — The Central Failure:** This is the paradigm's most fundamental and disqualifying weakness. A fine-tuned BERT model classifies based on *statistical patterns in text*. It has no internal representation of *why* the pattern is dangerous. It cannot verify whether the LOTO mentioned in the report was actually applied or bypassed. It cannot reason about whether a present ignition source is actually within the explosion radius of a detected HC release. This "correlation without causation" problem generates both false positives (flagging mentions of LOTO in training/educational contexts) and false negatives (missing novel descriptions of complete causal chains it hasn't seen in training).

**Alert Fatigue from Imprecise Precision:** Even well-tuned transformers produce precision scores of 0.50–0.65 when recall is pushed to  $\geq 0.90$ . In a high-volume reporting environment like OIL (potentially hundreds of reports per day across all field locations), a precision of 0.55 means nearly half of all alerts dispatched to safety officers are false positives. Sustained false positive rates above ~35% cause safety officers to begin dismissing alerts without review — the most dangerous operational failure mode.

**Data Leakage in Published Benchmarks:** As noted, most published transformer-based safety classification benchmarks use random train/test splits. When evaluated on a proper temporal split (train on older data, test on newer), performance typically drops by 5–12 percentage points — narrowing the gap with classical baselines.

**Black-Box Auditability Gap:** SHAP and LIME provide token-level attribution but do not provide safety-theoretic explanations. Telling a safety officer "the model flagged this because the word 'PTW' had high attribution" is not the same as "this report was flagged because it describes a hot work activity in proximity to hydrocarbon equipment without PTW authorization, which constitutes a failed administrative barrier over a chemical energy release hazard."

**GPU Dependency for Real-Time Inference:** Fine-tuned transformer inference requires GPU for sub-second latency targets. For remote OIL field locations with satellite connectivity only, cloud GPU inference introduces latency and

reliability dependencies.

**Catastrophic Forgetting on New SIF Categories:** When new SIF categories are introduced (e.g., a new hazard type emerges in operations), fine-tuned transformers must be retrained on combined old + new data to avoid catastrophic forgetting — a 4–8 week cycle that leaves the gap period without detection capability.

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### Paradigm 3: Graph & Ontology-Based Incident Modeling

#### ✓ Positive Remarks (Strengths)

**Explicit Causal Representation:** Knowledge graphs encode *named causal relationships* between safety entities — (H2S\_release) -[BLOCKED\_BY]-> (atmospheric\_testing) -[FAILED\_BECAUSE]-> (testing\_not\_conducted) — not just statistical co-occurrence. This is the only paradigm that represents *why* a situation is dangerous, not just *that* it historically correlates with dangerous outcomes.

**Zero Hallucination on Validated Subgraphs:** Within the domains covered by the knowledge graph, reasoning is formally correct. A SPARQL query over an HSE ontology produces only asserted or logically derived facts — it cannot "hallucinate" a causal relationship that is not in the graph.

**Regulatory Auditability:** OWL/RDF knowledge graphs can be mapped directly to regulatory frameworks (OSHA 29 CFR 1910, PSM/RMP requirements, API RP 75). Queries can demonstrate not only that a report indicates SIF potential but precisely which regulatory standard the detected barrier failure violates.

**Case-Based Reasoning for Novel Incidents:** Graph-based retrieval of structurally similar past incidents enables systematic comparison of a new near-miss against known SIF events — supporting the hypothesis that if a structurally identical past event resulted in a fatality, the current one has high SIF potential.

**Incremental Updateability:** New hazard types, equipment types, or regulatory requirements can be added to the ontology as new nodes and edges without retraining any ML model. The graph grows monotonically; it does not "forget."

#### ✗ Negative Remarks (Weaknesses & Bottlenecks)

**The NLP Bottleneck Problem:** Knowledge graphs require *structured input* — entity-populated slots matching the ontology schema. But raw incident reports are *unstructured natural language*. The NLP extraction layer (NER + relation extraction) that bridges these worlds is precisely the component where all the difficulty lives. A knowledge graph system's SIF detection quality is entirely bounded by the quality of its upstream NLP extraction — and that extraction is just as hard as the end classification task.

**Ontology Engineering is Expensive and Brittle:** Building a high-quality HSE ontology for O&G operations requires months of expert knowledge engineering effort. Domain experts must agree on formal definitions for hundreds of concepts. As operational practice evolves (new equipment types, new regulatory requirements, new work methods), the ontology must be manually updated — a process that is slow, expensive, and prone to inconsistency.

**Open-World Failure:** A knowledge graph operating under the Closed World Assumption (CWA) will produce "no information" for any incident type not present in the ontology. In practice, truly novel SIF scenarios — the ones most urgently needing detection — are precisely those not yet encoded. Open-world OWL reasoning partially addresses this but at significant computational cost.

**Scalability of Graph Traversal:** Complex ontological queries (multi-hop causal pathway traversal) on large graphs can be computationally intensive. A SPARQL query verifying the complete causal chain for an incoming report in real-time (< 500ms) requires carefully optimized graph indices and may not scale to high-volume report ingestion.

**Telegraphic Text Cannot be Reliably Parsed:** The NER and relation extraction models required to populate knowledge graph slots perform significantly worse on O&G's telegraphic, jargon-dense text style than on the formal scientific text they were typically trained on.

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## Paradigm 4: Multi-Modal / Vision-Text Systems

### ✓ Positive Remarks (Strengths)

**Corroborating Evidence from Independent Channels:** When text says "no fall arrest" and the attached photo shows a worker at height without harness, the two independent modalities *converge* on the same conclusion — dramatically increasing classification confidence. Conversely, when a report describes a serious hazard but the photo shows it has been corrected, the visual evidence can prevent a false positive alert.

**Handles Text Incompleteness:** Field workers often write incomplete or ambiguous reports while attaching a photograph that makes the situation unambiguous. Multi-modal systems exploit this complementarity; text-only systems cannot.

**Enables Proactive Hazard Detection:** CCTV and drone-based systems can detect SIF precursors *before* a report is even filed — transitioning from reactive (analyzing filed reports) to proactive (detecting unobserved conditions in real time).

**Spatiotemporal Fusion Adds Context:** Fusing the visual detection of a safety violation with the historical UA/UC text record for that specific location enables risk prioritization based on location-specific precursor history — a capability unique to the multi-modal paradigm.

### ✗ Negative Remarks (Weaknesses & Bottlenecks)

**Infrastructure Dependency:** High-quality CCTV and drone infrastructure in all relevant work areas is a significant capital investment, particularly for OIL's geographically dispersed onshore operations in Assam and Rajasthan. Many critical risk zones (underground, confined spaces, remote drill sites) cannot be continuously monitored by camera.

**Photo Availability is Highly Variable:** Safety observation cards frequently have no attached photo. Where photos exist, they vary enormously in quality, angle, and lighting. A model trained to extract safety-critical details from high-quality structured photos will underperform on the blurry, angled, poorly lit phone photographs typical of field safety observations.

**Modality Fusion Complexity:** Determining the optimal fusion strategy (early fusion, late fusion, cross-modal attention) for safety-specific text-image pairs is non-trivial and dataset-dependent. Fusion models require large multi-modal training datasets that are extremely difficult to assemble in a safety incident context.

**Privacy and Legal Constraints:** Continuous workplace CCTV surveillance introduces significant legal (industrial relations, data protection) and ethical complexities that may be prohibitive in unionized or regulated O&G workforce environments.

**Computer Vision Models Are Domain-Specific:** A PPE detection model trained on construction worker data will underperform on offshore process plant environments. Domain transfer requires retraining on O&G-specific visual data — a significant labeling burden.

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## Paradigm 5: Generative LLM Few-Shot / RAG Pipeline

### ✓ Positive Remarks (Strengths)

**Immediate Deployment Without Training Data:** LLM-based systems can be deployed with zero labeled training data. A well-engineered prompt producing SIF classification with chain-of-thought reasoning can be operational in days, not months — providing immediate value while the annotation pipeline is being built.

**Rich Causal Reasoning Output:** Unlike any classifier, an LLM can generate a complete chain-of-thought explanation: "This report describes [hot work activity] in proximity to [HC equipment], without [PTW authorization] or [atmospheric testing]. The failure of these two administrative barriers over a chemical energy source with a worker in the exposure zone constitutes a complete SIF causal chain..." This is qualitatively different from SHAP token attribution.

**RAG Grounds Reasoning in Real Precedents:** When an LLM retrieves structurally similar historical SIF events and explicitly reasons by analogy, its explanations are grounded in verified past incidents — reducing hallucination risk and providing safety officers with directly relevant comparison cases.

**Handles Novel Phrasing and Languages:** LLMs generalize to novel expressions of familiar hazardous situations and have multilingual capability without separate model training. Code-switching (Hinglish safety reports) is handled natively by multilingual LLMs.

**Low Labeling Cost for Prototyping:** Few-shot LLM classification enables rapid iteration on SIF criteria without annotation sprints. SMEs can evaluate the system quality by reviewing LLM reasoning traces rather than building training datasets.

### ✗ Negative Remarks (Weaknesses & Bottlenecks)

**Hallucination in Safety-Critical Context is Unacceptable:** LLMs can generate confident, plausible-sounding causal explanations for safety incidents that are factually incorrect. In a safety-critical system, a hallucinated false negative explanation — one that convinces a safety officer a genuine SIF precursor is benign — could directly contribute to a fatality. The absence of formal guarantees against hallucination is a disqualifying characteristic for standalone LLM deployment.

**Recall Instability Across Prompt Variations:** LLM classification performance is highly sensitive to prompt phrasing, token ordering, and context window position. Minor prompt changes can shift recall by 5–15 percentage points. Systematic prompt engineering mitigates this but does not eliminate instability. A fine-tuned discriminative classifier has more stable recall characteristics.

**Latency and Cost at Scale:** GPT-4 API calls with long context windows (RAG retrieval adds 2,000–5,000 tokens per call) at OIL's report ingestion volume could cost \$15,000–\$50,000 USD/month in API fees. More importantly, API latency of 3–8 seconds per report may be incompatible with real-time alerting requirements.

### **Data Privacy: Raw Incident Text Cannot Leave OIL's Perimeter**

Sending OIL's incident reports to a third-party LLM API (OpenAI, Anthropic) violates data governance requirements, exposes corporate-sensitive operational information, and may breach DPDP Act 2023 (India) provisions. On-premise LLM deployment (Llama 3, Mistral, Gemma) addresses this but requires significant infrastructure investment and produces lower performance than frontier models.

**No Structural Verification of Causal Completeness:** Even when an LLM reasons that a report "indicates SIF potential," it cannot formally verify whether the identified energy source actually exceeds the injury threshold, whether the identified barriers are *all* failed (versus merely mentioned), or whether the target is *in* the exposure zone. This verification requires domain-specific ontological knowledge that LLMs approximate but do not precisely represent.

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## Paradigm 6: Federated & Privacy-Preserving NLP

### ✓ Positive Remarks (Strengths)

**Enables Cross-Company Model Improvement Without Data Exposure:** Federated learning allows OIL to participate in an industry-wide SIF classifier improvement program without sharing a single incident report. Shared gradient updates improve the global model; each company's locally sensitive data remains within its perimeter.

**Formal Privacy Guarantees:** Differential privacy provides mathematical guarantees on the maximum information leakable about any individual incident record — a property no other paradigm offers.

**Contractor Data Integration Without Legal Risk:** OIL's contractors generate significant safety observation data that is currently siloed due to data ownership concerns. Federated learning enables contractor data to improve OIL's model without transferring custody of contractor-owned incident records.

**Regulatory Alignment:** Privacy-preserving frameworks align with DPDP Act 2023, GDPR (for European JV partners), and emerging HSE data governance standards from IOGP.

**✗ Negative Remarks (Weaknesses & Bottlenecks)**

**Performance Cost of Privacy Budget:** Differential privacy training with a strict privacy budget ( $\epsilon < 1$ ) typically degrades classification performance by 5–12 percentage points versus non-private training on the same data. At OIL's required Recall  $\geq 0.92$ , this performance cost may be prohibitive.

**Communication Overhead in Distributed Training:** Federated learning requires multiple rounds of gradient synchronization between local clients (individual field locations) and a central aggregator. For OIL's remote field locations with satellite or limited bandwidth connectivity, the communication overhead may make federated training cycles prohibitively slow.

**Non-IID Data Distribution:** Different OIL field locations (Assam vs. Rajasthan; onshore vs. pipeline operations) have different incident distributions. Federated averaging over non-identically-distributed (non-IID) local datasets produces globally suboptimal models that may underperform on any single location compared to a locally-trained model.

**Operational Complexity:** Federated learning infrastructure (secure aggregation server, client deployment across field locations, version control of model updates) adds significant DevOps complexity beyond the NLP pipeline itself — potentially doubling the implementation timeline.

Section 3: Structured Comparative Analysis Matrix

The following matrix evaluates all six paradigms across nine critical dimensions using a standardized 5-point scale (● = Inadequate, ● = Poor, ● = Moderate, ● = Good, ⚡ = Excellent):

| Dimension                  | Rule-Based<br>& Classical<br>NLP                     | Domain Fine-<br>Tuned<br>Transformers                    | Graph &<br>Ontology<br>Systems             | Multi-Modal<br>Vision-Text                                    | Generative<br>LLM / RAG                                     | Federated<br>& Privacy-<br>Preserving<br>NLP       |
|----------------------------|------------------------------------------------------|----------------------------------------------------------|--------------------------------------------|---------------------------------------------------------------|-------------------------------------------------------------|----------------------------------------------------|
| Primary NLP<br>Methodology | Keyword<br>regex + TF-<br>IDF feature<br>engineering | Pre-trained<br>LM fine-tuning<br>(BERT/RoBERTa<br>class) | OWL ontology<br>+ SPARQL +<br>KG traversal | CNN/ViT<br>visual encoder<br>+ BERT text<br>encoder,<br>fused | Prompt<br>engineering<br>+ vector<br>RAG + LLM<br>inference | FL gradient<br>aggregation<br>+ DP-SGD<br>training |
| SIF Recall<br>Achievement  | ● 0.72–0.80<br>(hard ceiling)                        | ● 0.88–0.94<br>(class-<br>weighted)                      | ● 0.65–0.75<br>(NLP<br>bottleneck)         | ● 0.85–0.92<br>(multi-modal<br>fusion)                        | ● 0.75–<br>0.85<br>(unstable)                               | ● 0.80–<br>0.88<br>(privacy<br>penalty)            |

| Dimension                                             | Rule-Based<br>& Classical<br>NLP              | Domain Fine-<br>Tuned<br>Transformers               | Graph &<br>Ontology<br>Systems                   | Multi-Modal<br>Vision-Text                     | Generative<br>LLM / RAG                   | Federated<br>& Privacy-<br>Preserving<br>NLP     |
|-------------------------------------------------------|-----------------------------------------------|-----------------------------------------------------|--------------------------------------------------|------------------------------------------------|-------------------------------------------|--------------------------------------------------|
| <b>Precision at<br/>Target Recall</b>                 | ● 0.35–0.50<br>at<br>Recall=0.85              | ● 0.50–0.65<br>at Recall=0.90                       | ● 0.68–0.78<br>(formal<br>verification)          | ● 0.62–0.75<br>(corroborating<br>evidence)     | ● 0.55–<br>0.70<br>(prompt-<br>dependent) | ● 0.48–<br>0.62<br>(privacy<br>penalty)          |
| <b>Causal<br/>Reasoning<br/>Capability</b>            | ● None —<br>pure<br>correlation               | ● Implicit<br>only — no<br>verified causal<br>chain | ⚡ Explicit<br>formal causal<br>representation    | ● Partial —<br>visual<br>corroboration<br>only | ● Generated<br>reasoning,<br>unverified   | ● Inherited<br>from base<br>model only           |
| <b>Telegraphic /<br/>O&amp;G Jargon<br/>Handling</b>  | ● Good<br>with explicit<br>acronym dict       | ● Good after<br>domain pre-<br>training             | ● Poor —<br>requires<br>structured<br>input      | ● Moderate<br>for text<br>component            | ● Good<br>with few-<br>shot<br>examples   | ● Depends on<br>base model                       |
| <b>Hindi/Assamese<br/>Code-Switch<br/>Support</b>     | ● None<br>without<br>translation              | ● XLM-<br>RoBERTa<br>native support                 | ● Ontology<br>is English-only                    | ● Partial<br>(text<br>component<br>only)       | ● Multilingual<br>LLMs<br>support         | ● Depends on<br>base model                       |
| <b>XAI &amp;<br/>Regulatory<br/>Auditability</b>      | ⚡ Fully<br>deterministic<br>& auditable       | ● SHAP/LIME<br>token<br>attribution<br>only         | ⚡ Formal<br>proof via<br>SPARQL trace            | ● Visual +<br>text<br>attribution              | ● Chain-<br>of-thought<br>(unverifiable)  | ● Inherited<br>from base<br>model                |
| <b>Alert Fatigue<br/>Risk</b>                         | ● High —<br>keyword<br>over-<br>triggering    | ● Moderate<br>— threshold-<br>manageable            | ● Low —<br>precise causal<br>verification        | ● Low —<br>dual-modality<br>confirmation       | ● Moderate —<br>prompt<br>instability     | ● Moderate                                       |
| <b>Zero/Low<br/>Training Data<br/>Feasibility</b>     | ⚡ Rule<br>system<br>needs no<br>training data | ● Minimum<br>500–1,000<br>labeled<br>examples       | ● Ontology-<br>based, no ML<br>labels needed     | ● Requires<br>labeled multi-<br>modal pairs    | ⚡ Zero-<br>shot / few-<br>shot capable    | ● Requires<br>distributed<br>labeled<br>datasets |
| <b>Implementation<br/>Feasibility for<br/>O&amp;G</b> | ⚡ Immediate,<br>no<br>infrastructure          | ● Moderate<br>— GPU,<br>annotation<br>sprint        | ● Complex<br>— ontology<br>engineering           | ● High infra<br>cost —<br>camera<br>coverage   | ● API<br>dependency<br>or on-prem<br>LLM  | ● Complex<br>distributed<br>infra                |
| <b>Real-Time<br/>Latency (&lt; 1<br/>second)</b>      | ⚡ < 10ms                                      | ● 200–<br>800ms (GPU)                               | ● 500ms–2s<br>(graph<br>traversal<br>complexity) | ● 1–3s<br>(fusion<br>pipeline)                 | ● 3–10s<br>(API +<br>retrieval)           | ● Inference<br>only post-<br>training            |

| Dimension                                            | Rule-Based<br>& Classical<br>NLP                     | Domain Fine-<br>Tuned<br>Transformers         | Graph &<br>Ontology<br>Systems            | Multi-Modal<br>Vision-Text                            | Generative<br>LLM / RAG                             | Federated<br>& Privacy-<br>Preserving<br>NLP |
|------------------------------------------------------|------------------------------------------------------|-----------------------------------------------|-------------------------------------------|-------------------------------------------------------|-----------------------------------------------------|----------------------------------------------|
| Sensitivity to<br>Concept Drift                      | ● Visible<br>failure mode                            | ● Silent<br>failure —<br>retraining<br>needed | ● Explicit<br>update via<br>graph editing | ● Moderate<br>(vision + text<br>drift)                | ● Few-shot<br>examples<br>updatable                 | ● Periodic<br>federation<br>rounds           |
| Scalability to<br>50,000+<br>Reports/Day             | ⚡ Trivially<br>scalable                              | ● Horizontally<br>scalable on<br>GPU cluster  | ● Graph DB<br>indexing<br>required        | ● Camera<br>infra + GPU<br>cost scales                | ● API cost<br>prohibitive<br>at scale               | ● Local<br>inference<br>scales               |
| O&G Domain<br>Data<br>Requirement                    | ● Depends<br>entirely on<br>expert rule<br>authoring | ● Pre-<br>training<br>reduces label<br>needs  | ● Ontology<br>must be<br>O&G-specific     | ● Multi-<br>modal O&G<br>pairs<br>extremely<br>scarce | ● Few-shot<br>examples<br>sufficient                | ● Each<br>client site<br>needs local<br>data |
| Regulatory<br>Compliance<br>(India DPDP Act<br>2023) | ⚡ On-<br>premise, no<br>data<br>exposure             | ⚡ On-<br>premise<br>model,<br>compliant       | ⚡ Graph<br>database on-<br>premise        | ● CCTV<br>compliance<br>complexity                    | ● API call<br>to third<br>party = non-<br>compliant | ⚡ Privacy-<br>by-design                      |

Matrix Synthesis: The White Space

Reading across all six paradigms, three critical capabilities are absent from every existing approach:

1. **No existing system can formally verify that a complete causal chain (Energy Source + Barrier Failure + Target Exposure) is present** — the necessary and sufficient conditions for a SIF precursor.
2. **No existing system combines interpretable causal verification with adaptive natural language understanding** at the linguistic complexity of O&G field reports.
3. **No existing system implements a closed-loop HITL calibration mechanism that updates both the NLP model and the causal knowledge graph** from safety officer feedback simultaneously.

These three gaps define the innovation space for the novel proposal in Section 4.

Section 4: The Novel Proposal — PRISM-SIF

4.1 Project Title & Unique Value Proposition

PRISM-SIF

Precursor Recognition through Integrated Safety Modeling for Serious Injury & Fatality Detection

**Tagline:** "From Pattern Detection to Causal Proof — The First Formally Verifiable SIF Alert System"

Unique Value Proposition

Every existing SIF detection system answers one question: **"Does this report text resemble past SIF reports?"**

PRISM-SIF answers a fundamentally different question: **"Does this report describe a situation in which an uncontrolled energy source, a failed critical barrier, and an exposed target are simultaneously present — thereby constituting a verified causal pathway to a Serious Injury or Fatality?"**

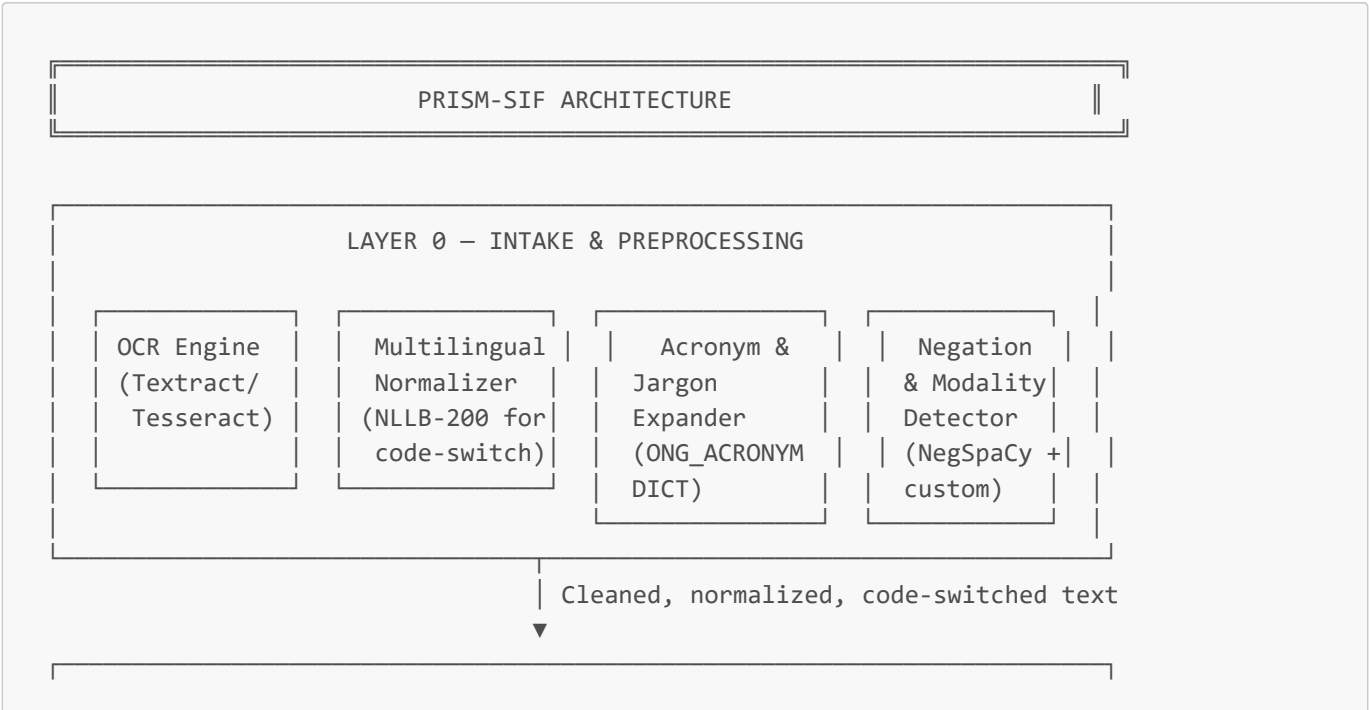
This distinction — between *statistical resemblance* and *causal verification* — is not incremental. It is architectural. PRISM-SIF does not just classify text; it reasons about safety in the way a trained HSE engineer reasons: by verifying whether the physical conditions necessary for a SIF event have been met, not merely whether the language resembles historical SIF language.

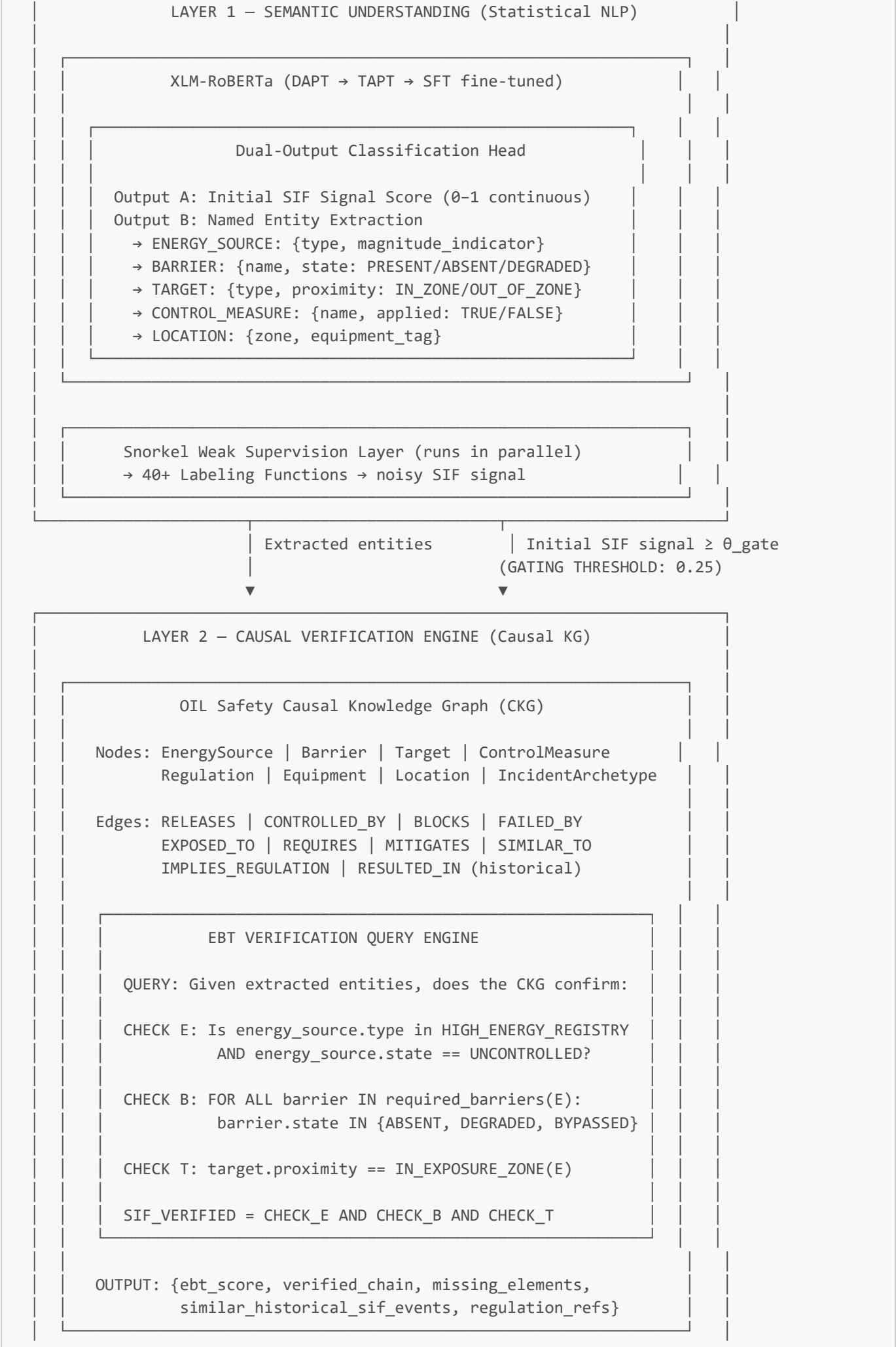
What makes PRISM-SIF distinct from every existing system:

| Capability                                                                   | All Existing Systems | PRISM-SIF |
|------------------------------------------------------------------------------|----------------------|-----------|
| Detects SIF keyword patterns                                                 | ✔ Yes                | ✔ Yes     |
| Understands semantic paraphrases of SIF conditions                           | ⚠ Transformers only  | ✔ Yes     |
| Formally verifies Energy + Barrier + Target (EBT) chain                      | ✖ None               | ✔ Yes     |
| Handles Hindi/Assamese code-switching                                        | ⚠ XLM-RoBERTa only   | ✔ Yes     |
| Generates safety-theoretic (not just token-level) explanations               | ✖ None               | ✔ Yes     |
| Updates from safety officer corrections simultaneously at NLP + Graph layers | ✖ None               | ✔ Yes     |
| Operates fully on-premise with no third-party API dependencies               | ⚠ Varies             | ✔ Yes     |

4.2 Hybrid Architecture Blueprint

PRISM-SIF is a **three-layer hybrid architecture** where each layer compensates for the other's fundamental limitations:







## Why This Tri-Layer Design Works

**Layer 1 alone fails** because statistical association  $\neq$  causal verification.

**Layer 2 alone fails** because the KG cannot parse unstructured natural language.

**Layer 3 alone fails** because LLMs hallucinate and cannot formally verify causal completeness.

### Together:

- Layer 1 extracts *what is described* in natural language with semantic understanding
- Layer 2 verifies *whether what is described constitutes a complete SIF causal chain* using formal domain knowledge
- Layer 3 explains *why it is dangerous* in terms a safety officer can use for immediate decision-making

Each layer checks the others. The system only produces a high-confidence SIF alert when all three layers agree.

---

### 4.3 Causal SIF Pathway Inference — The EBT Verification Mechanism

The Energy-Barrier-Target (EBT) model is the formal causal framework at PRISM-SIF's core. It operationalizes the Swiss Cheese Model, Bow-Tie methodology, and STAMP/STPA control-theoretic principles into a computationally executable verification check.

#### EBT Theoretical Foundation

The model asserts that a SIF event requires — and only requires — three conditions to be simultaneously true:

```
SIF_CAUSAL_CHAIN_COMPLETE ←
  CONDITION_E: Uncontrolled high-energy hazard is present
  AND
  CONDITION_B: At least one critical barrier in the required barrier set has failed
  AND
  CONDITION_T: A human target is within the exposure zone of the energy source
```

This formalization has direct roots in Haddon's Energy Release Theory, Reason's Swiss Cheese Model, and Leveson's STAMP framework — three independently developed theoretical traditions that converge on the same conclusion.

#### EBT Causal Knowledge Graph Schema

```
# OWL/RDF Schema excerpt for the OIL Safety CKG

:EnergySource a owl:Class ;
  rdfs:label "HazardousEnergySource" ;
  rdfs:subClassOf [
    a owl:Restriction ;
    owl:onProperty :hasEnergyType ;
    owl:someValuesFrom :EnergyType
  ] .

:EnergyType a owl:Class ;
  owl:equivalentClass [
    a owl:Class ;
    owl:oneOf (:Chemical_Hydrocarbon :Chemical_Toxic :Electrical
      :Pressure_Mechanical :Gravitational :Thermal :Kinetic)
  ] .

:Barrier a owl:Class ;
  rdfs:label "SafetyBarrier" ;
  rdfs:subClassOf [
    a owl:Restriction ;
    owl:onProperty :hasBarrierType ;
    owl:someValuesFrom :BarrierType
  ] .

:BarrierType a owl:Class ;
  owl:equivalentClass [
    a owl:Class ;
```

```

        owl:oneOf (:Administrative_PTW :Administrative_JSA :Administrative_LOTO
                    :Engineering_ESD :Engineering_PRD :Engineering_Guarding
                    :PPE_BA :PPE_FallArrest :Procedural_GasTest)
    ] .

:BarrierState a owl:Class ;
    owl:equivalentClass [
        a owl:Class ;
        owl:oneOf (:PRESENT_EFFECTIVE :DEGRADED :ABSENT :BYPASSED :UNKNOWN)
    ] .

# Critical relation: required barrier sets per energy type
:Chemical_Hydrocarbon :requiresBarrier :Administrative_PTW ,
                        :Procedural_GasTest ,
                        :Engineering_ESD ,
                        :PPE_BA .

:Electrical :requiresBarrier :Administrative_LOTO ,
                            :Engineering_Guarding ,
                            :PPE_Insulation .

```

## EBT Verification Logic (Python/SPARQL)

```

class EBTVerifier:
    """
    Formally verifies whether extracted NLP entities constitute
    a complete SIF causal chain using the Causal Knowledge Graph.
    """

    def __init__(self, causal_kg: Graph, energy_registry: dict):
        self.kg = causal_kg
        self.energy_registry = energy_registry

    def verify_ebt_chain(self, nlp_output: dict) -> EBTResult:
        """
        nlp_output: {
            'energy_sources': [{'type': str, 'state': str, 'magnitude': str}],
            'barriers': [{'name': str, 'state': str}],
            'targets': [{'type': str, 'proximity': str}],
            'controls': [{'name': str, 'applied': bool}]
        }
        """
        # — CHECK E: Uncontrolled High-Energy Source —————
        energy_check = self._verify_energy_condition(nlp_output['energy_sources'])

        # — CHECK B: Critical Barrier Failure —————
        # For each detected energy source, query KG for required barriers
        # Then verify at least one is in ABSENT/BYPASSED/DEGRADED state
        barrier_check = self._verify_barrier_failure(
            energy_sources=nlp_output['energy_sources'],
            detected_barriers=nlp_output['barriers']
        )

        # — CHECK T: Target in Exposure Zone —————
        target_check = self._verify_target_exposure(nlp_output['targets'])

```

```

# — RETRIEVE ANALOGOUS HISTORICAL EVENTS —————
similar_events = self._retrieve_similar_sif_events(
    energy_type=energy_check.energy_type,
    failed_barrier_types=[b.name for b in barrier_check.failed_barriers]
)

# — COMPUTE COMPOSITE EBT SCORE —————
# All three checks needed for full verification
# Partial matches produce intermediate scores (for graduated alerting)
ebt_score = self._compute_ebt_score(energy_check, barrier_check, target_check)

return EBResult(
    chain_complete=(energy_check.verified and
                    barrier_check.verified and
                    target_check.verified),
    ebt_score=ebt_score,
    energy_verification=energy_check,
    barrier_verification=barrier_check,
    target_verification=target_check,
    similar_historical_events=similar_events,
    missing_elements=self._identify_missing_elements(
        energy_check, barrier_check, target_check
    )
)

def _verify_barrier_failure(self, energy_sources, detected_barriers) -> BarrierCheck:
    """
    Key innovation: For each energy source, retrieve the REQUIRED barrier set
    from the CKG. Then check whether the NLP-detected barrier states indicate
    failure. This prevents false positives where barriers are mentioned but
    confirmed as applied.
    """
    sparql_query = """
PREFIX sif: <http://oil.in/safety/ontology#>

SELECT ?barrier ?barrier_type WHERE {
    ?energy_source a sif:EnergySource ;
        sif:hasEnergyType sif:%s ;
        sif:requiresBarrier ?barrier .
    ?barrier a sif:Barrier ;
        sif:hasBarrierType ?barrier_type .
}
""" % energy_sources[0]['type']

required_barriers = set(self.kg.query(sparql_query))
failed_barriers = []

for req_barrier in required_barriers:
    # Check if this barrier was detected by NLP layer
    detected = self._find_in_detected(req_barrier, detected_barriers)
    if detected is None:
        # Barrier not mentioned = assume unknown state (partial flag)
        failed_barriers.append(BarrierFailure(req_barrier, 'NOT_MENTIONED'))
    elif detected['state'] in ['ABSENT', 'BYPASSED', 'DEGRADED']:
        # Barrier explicitly failed
        failed_barriers.append(BarrierFailure(req_barrier, detected['state']))
    # If detected['state'] == 'PRESENT' → barrier is functioning, no failure

```

```
# Critical: barrier_failure verified only if at least ONE is confirmed failed
# Not just absent from text – requires NEGATIVE evidence, not merely missing
confirmed_failures = [b for b in failed_barriers
                      if b.failure_mode in ('ABSENT', 'BYPASSED', 'DEGRADED')]

return BarrierCheck(
    verified=len(confirmed_failures) >= 1,
    failed_barriers=confirmed_failures,
    unconfirmed_absent=len(failed_barriers) - len(confirmed_failures)
)
```

Graduated Alert Tiers Based on EBT Completeness

A critical operational design decision: not all SIF alerts are equal. PRISM-SIF uses the EBT check's partial results to produce **graduated alert tiers** — reducing alert fatigue while ensuring no complete chain is missed:

| EBT Checks Verified                               | Alert Tier              | Action                                                                                                  |
|---------------------------------------------------|-------------------------|---------------------------------------------------------------------------------------------------------|
| E  + B  + T  (Complete chain)                     | <b>TIER 1: CRITICAL</b> | Immediate SMS + phone call to Safety Officer + Area HSE Lead. Auto-create Stop Work investigation task. |
| E  + B  + T  (Target exposure uncertain)          | <b>TIER 2: HIGH</b>     | Push notification to Safety Officer. Flag for priority review within 30 minutes.                        |
| E  + B  + T  (Energy confirmed, barriers unclear) | <b>TIER 3: ELEVATED</b> | Dashboard flag. Safety Officer review within 2 hours. Request location verification.                    |
| E  (High SIF signal from L1, EBT incomplete)      | <b>TIER 4: WATCH</b>    | Logged to SIF watchlist. Aggregated for weekly pattern analysis. No individual alert.                   |
| No SIF signal                                     | No alert                | Normal EHS workflow.                                                                                    |

This tiered structure is directly analogous to the DEFCON alert system — calibrated response proportional to verified threat level, not a binary on/off alarm.

4.4 Closed-Loop Human-in-the-Loop (HITL) Workflow

The HITL system is not an afterthought — it is the mechanism by which PRISM-SIF improves continuously from the accumulated judgment of domain experts without requiring expensive retraining cycles.

HITL Design Principles

- Minimum Friction:** A safety officer must be able to review and respond to an alert in under 90 seconds on a mobile device, under field conditions (gloves, bright sunlight, time pressure).
- Structured Feedback:** Responses must be structured (not free-text) to enable automated model updating. But optional free-text notes capture information that structured fields cannot.
- Bidirectional Learning:** Safety officer decisions update *both* the NLP model (via active learning) *and* the Causal Knowledge Graph (via graph refinement).
- Federated Aggregation:** Corrections from multiple field locations (Duliajan, Jorhat, Rajasthan) are aggregated centrally to improve the global model, while location-specific signal drives local threshold calibration.
- Explainability First:** Safety officers see the EBT reasoning chain *before* making a decision — they are validating an argument, not overriding a black box.

Mobile HITL Interface — Three-Screen Workflow

SCREEN 1: ALERT SUMMARY

 TIER 1 SIF ALERT

Report: OIL-2024-08-29-0847

Location: DULIAJAN NORTH FIELD

Filed: 08:47 by FIELD OPS

"HW carried out nr HC equip  
w/o PTW. No gas test done."

SIF Probability: 0.91 (HIGH)

EBT Chain:  COMPLETE

[View Reasoning]

[Quick Action]

Tap "View Reasoning"

SCREEN 2: EBT CAUSAL CHAIN

 ENERGY SOURCE

Type: Chemical Hydrocarbon

Indicator: "nr HC equip"

 High-energy threshold met

 FAILED BARRIERS

 PTW: "w/o PTW" → ABSENT

 Gas Test: "No gas test" → ABSENT

 TARGET EXPOSED

 Worker conducting hot work  
in proximity to HC equip

 SIMILAR INCIDENTS

- CSB Texas City 2005 (fatal)
- OIL-2021-JORHAT-HW-03

Regulation: OIL SMP-HW-007  
OISD-STD-105

[Confirm SIF]

[Dismiss]

[Edit]

Tap "Confirm SIF"

SCREEN 3: STRUCTURED FEEDBACK

Quick confirmation:

Was EBT assessment correct?

- o Yes, all 3 elements present
- o Energy correct, barriers OK

```

o Completely wrong (dismiss)

Action taken:
o Stop Work Order issued
o Verbal warning given
o Area barricaded
o Investigation launched

Corrected energy type? [v]
Notes: _____

[Submit & Close] <- 90 sec total

```

## HITL Learning Architecture

```

class HITLCalibratorService:
    """
    Processes safety officer feedback and triggers dual-layer model updates:
    1. NLP Model update via active learning (Label Studio → training pipeline)
    2. Causal Knowledge Graph update via ontology editor API
    """

    def __init__(self, nlp_trainer: NLPTrainer, ckg_editor: CKGEditor):
        self.nlp_trainer = nlp_trainer
        self.ckg_editor = ckg_editor
        self.feedback_buffer = []          # Collect feedback before batch update
        self.update_threshold = 50        # Trigger update after 50 confirmations

    def process_feedback(self, alert_id: str, feedback: SOFeedback):
        """
        SOFeedback schema:
        {
            'decision': 'CONFIRM_SIF' | 'DISMISS' | 'PARTIAL_CORRECT',
            'ebt_corrections': {
                'energy_correct': bool,
                'barrier_correct': bool,
                'target_correct': bool,
                'corrected_energy_type': str | None,
                'missed_barriers': [str] | None
            },
            'action_taken': str,
            'notes': str | None
        }
        """
        # Retrieve original report and PRISM output for this alert
        original = self.retrieve_alert_context(alert_id)

        # — ROUTE 1: NLP TRAINING UPDATE —————
        if feedback.decision in ('CONFIRM_SIF', 'DISMISS'):
            # Create a high-quality labeled training example from SO judgment
            training_example = TrainingExample(
                text=original.report_text,
                label=1 if feedback.decision == 'CONFIRM_SIF' else 0,
                entity_corrections=feedback.ebt_corrections,

```

```

        source='SO_CONFIRMED',          # Highest-quality label source
        annotator_confidence='HIGH'
    )
    self.feedback_buffer.append(training_example)

# — ROUTE 2: CAUSAL KG UPDATE —————
if feedback.ebt_corrections and feedback.ebt_corrections.get('missed_barriers'):
    # Safety officer identified barriers the NLP missed
    # → Update KG: add this barrier type to required_barriers for this energy
type
    for missed_barrier in feedback.ebt_corrections['missed_barriers']:
        self.ckg_editor.add_required_barrier(
            energy_type=original.ebt_result.energy_verification.energy_type,
            barrier_type=missed_barrier,
            evidence_source=f"SO_FEEDBACK_{alert_id}"
        )

    if feedback.ebt_corrections and
feedback.ebt_corrections.get('corrected_energy_type'):
    # Safety officer corrected the energy type classification
    # → Update KG: strengthen the corrected type's NLP→KG mapping
    self.ckg_editor.update_energy_type_mapping(
        nlp_phrase=original.nlp_output.energy_sources[0].indicator_phrase,
        correct_energy_type=feedback.ebt_corrections['corrected_energy_type']
    )

# — TRIGGER BATCH RETRAINING IF THRESHOLD MET —————
if len(self.feedback_buffer) >= self.update_threshold:
    self._trigger_nlp_update()
    self.feedback_buffer = []

def _trigger_nlp_update(self):
    """
    Efficient model update strategy:
    - Full retraining: monthly (if >200 new confirmed examples)
    - Threshold recalibration: weekly (using all SO feedback)
    - Labeling function update: immediate (Snorkel LF recalibration)
    """
    # Step 1: Always update decision threshold first (cheapest, highest impact)
    self.nlp_trainer.recalibrate_threshold(
        new_confirmed_examples=self.feedback_buffer,
        target_recall=0.92
    )

    # Step 2: Update Snorkel labeling functions weights
    self.nlp_trainer.update_label_model(
        new_feedback=self.feedback_buffer
    )

    # Step 3: Queue full fine-tuning if enough new examples
    if len([e for e in self.feedback_buffer
            if e.source == 'SO_CONFIRMED']) >= 50:
        self.nlp_trainer.schedule_fine_tuning(
            new_examples=self.feedback_buffer,
            retrain_mode='INCREMENTAL' # LoRA fine-tuning to prevent catastrophic
forgetting
        )

```

## LoRA-Based Incremental Fine-Tuning for Continuous Learning

A critical engineering decision: full fine-tuning every 50 new examples is computationally prohibitive. Instead, use **Low-Rank Adaptation (LoRA)** for parameter-efficient incremental updates:

```
from peft import LoraConfig, get_peft_model

lora_config = LoraConfig(
    task_type="SEQ_CLS",
    r=8,                # Low-rank decomposition dimension
    lora_alpha=32,      # Scaling factor
    target_modules=["query", "value"], # Apply LoRA only to attention layers
    lora_dropout=0.1,
    bias="none"
)

# LoRA update uses only ~0.3% of original parameter count
# Training on 50 new examples takes ~8 minutes on a single GPU
# No catastrophic forgetting of previous knowledge
peft_model = get_peft_model(base_xlm_roberta, lora_config)
```

## Drift Monitoring & Escalation Protocol

WEEKLY AUTOMATED DRIFT REPORT (generated every Monday, sent to HSE IT Lead):

PRISM-SIF Weekly Performance Dashboard – Week 34, 2024

Alert Volume: 247 alerts generated this week

Tier 1 (Critical): 18 | Tier 2 (High): 41

Tier 3 (Elevated): 88 | Tier 4 (Watch): 100

S0 Decision Breakdown (for 127 reviewed alerts):

✅ Confirmed SIF: 79 (62%) → Precision = 0.62

❌ Dismissed: 38 (30%) → False Positive rate

⚠️ Partial correct: 10 (8%)

RECALL ESTIMATE (this week):

Recall on S0-reviewed subset: 0.91 ✅ (target: ≥0.92)

PSI (feature drift index): 0.08 ✅ (threshold: 0.10)

CKG Updates this week:

New barrier mappings added: 3

Energy type corrections: 1

New historical SIF event indexed: 2

⚠️ DRIFT WARNING: "grinding near HC line" phrase cluster showing 23% higher frequency than 4-week baseline. Recommend SME review of Tier 4 Watch items in JORHAT PROCESSING AREA.

LoRA fine-tune queued: YES (63 confirmed examples)

ETA: Sunday 02:00 AM (off-peak maintenance window)

### 4.5 Step-by-Step Practical Implementation Roadmap

#### Phase 0 — Foundation (Weeks 1–4)

**Objective:** Establish governance, data access, and infrastructure before writing a single line of NLP code.

| Task                                                                                 | Owner                | Deliverable                                                                 | Risk                                 |
|--------------------------------------------------------------------------------------|----------------------|-----------------------------------------------------------------------------|--------------------------------------|
| Establish Data Classification Committee (EHS + Legal + IT Security)                  | HSE Head + CIO       | Approved data governance framework                                          | Medium — needs executive sponsorship |
| Sign data extraction agreements with all EHS system vendors (SAP, Enablon)           | IT Procurement       | Signed API access agreements                                                | Low                                  |
| Conduct OIL field report language audit (30 sample reports from each field location) | NLP Lead + Field HSE | Linguistic profile report identifying code-switch frequency, jargon density | Low                                  |
| Select and procure annotation platform (Label Studio recommended for on-premise)     | IT + NLP Lead        | Running annotation instance                                                 | Low                                  |
| Define EBT ontology scope with HSE SMEs: which energy types, which barrier sets      | HSE SMEs + NLP Lead  | CKG ontology schema v0.1 (OWL file)                                         | High — requires significant SME time |
| Deploy baseline monitoring: DVC, MLflow, Great Expectations                          | AI Engineering       | MLOps infrastructure operational                                            | Low                                  |

**Phase 0 Gate Criteria:** Data governance approved, annotation platform live, ontology schema v0.1 signed off by ≥2 HSE SMEs.

#### Phase 1 — Data Pipeline & Initial Labeling (Weeks 5–12)

**Objective:** Build the clean, anonymized dataset and establish Snorkel weak supervision baseline.

| Task                                                                         | Owner          | Deliverable                                               |
|------------------------------------------------------------------------------|----------------|-----------------------------------------------------------|
| Extract 5-year historical UA/UC and Near-Miss reports from SAP EHS / OIL SMS | AI Engineering | Raw dataset: ~20,000–50,000 reports                       |
| Run Presidio PII anonymization pipeline                                      | AI Engineering | Anonymized dataset in Delta Lake                          |
| Build OCR pipeline for handwritten safety cards                              | AI Engineering | Digitized card dataset (additional ~5,000–10,000 records) |
| Collect and process all Tier 1 external datasets (OSHA SIR, BSEE, CSB PDFs)  | NLP Lead       | External pre-training corpus (~200,000 documents)         |

| Task                                                                                 | Owner               | Deliverable                                                      |
|--------------------------------------------------------------------------------------|---------------------|------------------------------------------------------------------|
| Author 40+ Snorkel Labeling Functions across all 9 SIF energy categories             | NLP Lead + HSE SMEs | Snorkel LF library; weak labels for ~70% of dataset              |
| Run inter-annotator agreement study (200 reports, 2 SMEs each) to calibrate taxonomy | HSE SMEs            | IAA Cohen's Kappa target $\geq 0.75$ ; calibrated labeling guide |
| Build ONG_ACRONYM_DICT (target: 200+ acronyms with O&G HSE domain coverage)          | HSE SMEs            | Validated acronym expansion dictionary                           |

**Phase 1 Gate Criteria:**  $\geq 15,000$  weak-labeled reports in Delta Lake; IAA  $\kappa \geq 0.75$  on 200-report calibration set; Snorkel LF coverage  $> 65\%$ .

## Phase 2 — Baseline Modeling & CKG Build (Weeks 13–20)

**Objective:** Deploy interpretable baseline and begin CKG construction in parallel.

| Task                                                                          | Owner                | Deliverable                                              |
|-------------------------------------------------------------------------------|----------------------|----------------------------------------------------------|
| Train TF-IDF + SVM / Logistic Regression baseline on temporally split dataset | NLP Lead             | Baseline model with evaluated recall, F2, PR-AUC         |
| Deploy keyword rule system (immediate quick-win for field teams)              | AI Engineering       | Rule-based SIF screener in production (shadow mode)      |
| Run DAPT on XLM-RoBERTa using OSHA + BSEE + CSB external corpus               | AI Engineering (GPU) | Domain-adapted language model checkpoint                 |
| Run TAPT on OIL's internal anonymized reports                                 | AI Engineering (GPU) | OIL-specific TAPT checkpoint                             |
| Build CKG v0.1: ingest and encode all CSB + BSEE + OIL confirmed SIF events   | Knowledge Engineer   | Operational CKG with ~500 SIF incident nodes             |
| Implement NER training for ENERGY_SOURCE, BARRIER, TARGET, CONTROL entities   | NLP Lead             | Custom spaCy NER model ( $F1 \geq 0.78$ on held-out set) |
| Set up SPARQL-based EBT verification query engine                             | Knowledge Engineer   | EBT verifier prototype (unit-tested)                     |

**Phase 2 Gate Criteria:** Baseline recall  $\geq 0.80$  on temporal test split; CKG operational with EBT verification returning results; NER  $F1 \geq 0.78$ .

## Phase 3 — Active Learning Annotation Sprint (Weeks 21–28)

**Objective:** Build gold-standard annotated training set to 2,000+ examples using active learning.

| Task                                                                             | Owner               | Deliverable                                            |
|----------------------------------------------------------------------------------|---------------------|--------------------------------------------------------|
| Run active learning iteration 1–3 (500 seed → uncertainty sample → SME annotate) | NLP Lead + HSE SMEs | 500 SME-confirmed gold labels (iterations 1–3)         |
| LLM-assisted pre-labeling for active learning samples (Llama-3 on-premise)       | AI Engineering      | Pre-labeled drafts reducing SME cognitive load by ~60% |

| Task                                                                               | Owner                          | Deliverable                                                            |
|------------------------------------------------------------------------------------|--------------------------------|------------------------------------------------------------------------|
| SME annotation sprint: 3 HSE experts × 8 hours/week × 6 weeks                      | HSE SMEs (resource commitment) | 1,500 additional gold labels (iterations 4–8)                          |
| Validate annotation quality: Escalation_Flag review, pathway rationale audit       | HSE Lead                       | Annotation QA report; total confirmed dataset: ~2,000 labeled examples |
| Expand CKG from annotated reports: extract confirmed EBT chains as new graph edges | Knowledge Engineer + NLP Lead  | CKG v0.5 with ~2,000 confirmed SIF chains indexed                      |

**Phase 3 Gate Criteria:**  $\geq 2,000$  gold-labeled examples ( $\kappa \geq 0.75$ ); CKG populated with all confirmed SIF chains; Active learning performance plateau reached.

#### Phase 4 — Transformer Fine-tuning & System Integration (Weeks 29–36)

**Objective:** Fine-tune the full PRISM-SIF tri-layer system and integrate with OIL's EHS infrastructure.

| Task                                                                           | Owner                         | Deliverable                                                                     |
|--------------------------------------------------------------------------------|-------------------------------|---------------------------------------------------------------------------------|
| Supervised fine-tuning of XLM-RoBERTa on gold-labeled dataset (temporal split) | AI Engineering (GPU)          | SIF classifier: target Recall $\geq 0.92$ , F2 $\geq 0.75$ on temporal test set |
| Fine-tune NER component jointly with classifier (multi-task learning)          | AI Engineering                | Joint NER + classification model                                                |
| Deploy on-premise Llama-3-70B or Mistral-Large-2 with structured output mode   | AI Engineering (server infra) | Local LLM explanation engine (no API dependency)                                |
| Build HITL mobile interface (React Native progressive web app)                 | Frontend Engineer             | Safety Officer mobile alert review app (3-screen workflow)                      |
| Integrate PRISM-SIF API with SAP EHS webhook (new report → auto-analyze)       | AI Engineering + IT SAP team  | End-to-end pipeline: report → analysis → alert in < 2 seconds                   |
| Implement HITL calibrator service (feedback → NLP update → CKG update)         | AI Engineering                | Closed-loop learning pipeline operational                                       |
| SHAP + EBT explanation fusion: generate safety-theoretic alert narratives      | NLP Lead                      | Alert narrative generator tested against 50 SO review cases                     |
| Security audit of full pipeline (data at rest, data in transit, model access)  | IT Security                   | Signed security compliance report                                               |

**Phase 4 Gate Criteria:** Full PRISM-SIF system passes integration test; Recall  $\geq 0.92$  and F2  $\geq 0.75$  on held-out temporal test set; end-to-end latency < 2 seconds; security audit passed.

#### Phase 5 — Pilot Deployment (Weeks 37–44)

**Objective:** Shadow deployment at one OIL field location with full SO feedback collection.

| Task                                                                          | Owner    | Deliverable                                      |
|-------------------------------------------------------------------------------|----------|--------------------------------------------------|
| Select pilot location: recommend Duliajan North Field (highest report volume) | HSE Head | Pilot location confirmed; local HSE team briefed |

| Task                                                                                                                                     | Owner                           | Deliverable                                            |
|------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|--------------------------------------------------------|
| Shadow mode deployment: PRISM-SIF runs in parallel with existing process; no action taken on PRISM alerts without SO manual confirmation | AI Engineering + Pilot HSE Team | Parallel alert log: PRISM vs. existing flagging system |
| SO onboarding: 2-hour training on HITL interface and EBT reasoning review                                                                | HSE Lead                        | Trained SO team at pilot location (5–8 officers)       |
| Weekly performance calibration meetings (AI team + HSE SMEs, 45 min/week)                                                                | NLP Lead + HSE Lead             | Weekly calibration report with threshold adjustments   |
| Collect 8 weeks of SO feedback data                                                                                                      | Pilot HSE Team                  | ≥500 structured feedback events                        |
| Measure: Alert Conversion Rate, MTTA, SO Trust Score (survey), False Negative Rate                                                       | NLP Lead + HSE Head             | Pilot evaluation report                                |

**Phase 5 Gate Criteria:** Alert Conversion Rate (precision in operational context)  $\geq 0.60$ ; MTTA < 5 minutes; SO Trust Score > 3.5/5.0 on post-pilot survey; Zero confirmed SIF events missed (Recall on pilot-period incidents = 1.0).

## Phase 6 — Full Rollout & Continuous Improvement (Weeks 45–60+)

**Objective:** Deploy PRISM-SIF across all OIL field locations with federated location-specific calibration.

| Task                                                                                                                 | Owner                                 | Deliverable                                                            |
|----------------------------------------------------------------------------------------------------------------------|---------------------------------------|------------------------------------------------------------------------|
| Rollout to remaining OIL locations (Jorhat, Rajasthan, pipeline operations)                                          | AI Engineering + Field HSE Teams      | Multi-location deployment with location-specific threshold calibration |
| Establish monthly LoRA fine-tuning cadence (off-peak Sunday maintenance window)                                      | AI Engineering                        | Automated retraining pipeline (MLflow + DVC managed)                   |
| Monthly CKG expansion: new confirmed SIF events, regulatory updates, new hazard types                                | Knowledge Engineer                    | CKG version-controlled update log                                      |
| Quarterly SME calibration workshop: review accumulated HITL data, update labeling functions                          | HSE SMEs + NLP Lead                   | Updated Snorkel LF library; revised annotation guidelines              |
| Annual red-team exercise: HSE experts attempt to construct SIF precursor descriptions that evade PRISM-SIF detection | External HSE consultant               | Adversarial test report; model hardening patch                         |
| Extend CKG to include contractor safety data via federated learning protocol                                         | AI Engineering + Contractor EHS leads | Federated CKG update mechanism                                         |
| Publish external benchmark report (using anonymized performance data)                                                | Research Lead                         | Industry peer publication / IOGP technical report contribution         |

## 4.6 PRISM-SIF: Summary Innovation Claims

| Innovation Dimension     | PRISM-SIF's Contribution                                                                                                                                                                                                        |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Core Theoretical Advance | First system to implement formal Energy-Barrier-Target (EBT) causal verification as a computational safety check, bridging Haddon's Energy Theory, Reason's Swiss Cheese Model, and Leveson's STAMP into an executable pipeline |
| Architectural Novelty    | Tri-layer hybrid (Statistical NLP → Causal KG Verification → LLM Explanation) where each layer compensates for the others' fundamental limitations — no existing O&G system implements all three                                |
| Operational Advance      | Graduated 4-tier alert system driven by EBT completeness score, directly reducing alert fatigue compared to binary SIF/NonSIF classification                                                                                    |
| Linguistic Advance       | Native code-switching support (Hindi/Assamese/English) via XLM-RoBERTa, specifically calibrated for OIL's field reporting population                                                                                            |
| HITL Advance             | Bidirectional feedback loop updating both NLP model (via LoRA fine-tuning) and Causal Knowledge Graph simultaneously from safety officer corrections                                                                            |
| Privacy Advance          | Fully on-premise architecture (no third-party API) with LoRA parameter-efficient updates enabling future federated learning across contractor sites                                                                             |
| Explainability Advance   | EBT-grounded safety-theoretic explanation (not SHAP token attribution) that maps directly to the reasoning a trained HSE engineer would use                                                                                     |
| Organizational Advance   | HITL workflow designed for field conditions (mobile-first, < 90 seconds, gloved-hand compatible) rather than desktop HSE system UX                                                                                              |

## Appendix C: PRISM-SIF vs. Prior Synthesis (v1.0) — Upgrade Map

The v1.0 Research Synthesis (predecessor document) established the foundational data lifecycle. PRISM-SIF builds on and extends that foundation:

| v1.0 Component           | PRISM-SIF Extension                                                                                       |
|--------------------------|-----------------------------------------------------------------------------------------------------------|
| TF-IDF + SVM baseline    | → Retained as Tier-4 fallback in ensemble; CausalKG-informed feature selection improves TF-IDF lexicon    |
| XLM-RoBERTa fine-tuning  | → Extended with multi-task NER head (Entity extraction for EBT populator); LoRA for incremental updates   |
| Snorkel weak supervision | → LF library expanded to 40+ functions; LF outputs now also populate CKG gating conditions                |
| Threshold calibration    | → Extended to per-tier thresholds (4 tiers); SO feedback directly recalibrates tier boundaries            |
| SHAP/LIME explainability | → Augmented with EBT causal narrative (safety-theoretic, not just token attribution)                      |
| Alert dispatch           | → Extended to 4-tier graduated system with specific action protocols per tier                             |
| Monthly retraining       | → LoRA incremental updates (8 min/cycle) replace full fine-tuning; triggered at 50 new confirmed examples |

*PRISM-SIF Innovation Synthesis Version 1.0*

*Prepared as Successor to SIF Precursor NLP Research Synthesis v1.0*

*Classification: Internal Research Use | OIL SIF Precursor AI/NLP Engine Project*